

Seven findings across eight pages, measured at 360, 390, 430 and 820 pixels. Six fixed, one left open with a stated reason. Three of the seven were not mobile defects — opening the site on a phone is what prompted the audit that found them.

01 Findings

#	SYMPTOM	MEASUREMENT	CHANGE	VERIFIED
1	A reader on a phone reaches that chart and can't read it completely	4.2px → 12.1px	Both SVGs re-typeset, 960×520 → 360×456 and 360×150; labels moved above the bars	Fresh clone at 6a46779, four widths
2	Nav links too small to tap; two rows touching	height 20px → 33px, row pitch 20px → 41px	style.css — nav to flex-wrap with a row gap, padding on the links	WCAG 2.2 SC 2.5.8 minimum is 24px
3	Caption prose set in mono	Plex Mono 12px → Plex Sans 14px	style.css — figcaption rule	Identity kit rule: never prose in mono
4	Two feature labels clipped off the canvas at every width, desktop included. The visible string rd_count_tier_1000-2000 is not a column name in the repo, so a reader checking the figure against the source cannot reproduce it	start x=-17 → no negative x	Fixed by the re-typeset in finding 1	No label starts left of 0
5	Alt text named 6 of 12 features	6 → all twelve, summarised	cases.html — alt text rewritten	Found by counting, not by looking
6	The caption describes a file it has no connection to, so the file changed and the sentence did not — it still read “recoloured” after both charts had been re-typeset	repo 960×520 vs site 360×456	Captions corrected; source paths linked, matching every other repo reference on the site	Both paths resolve at cbcd11a; pushed at e52976a
7	Body links 19px, under the 24px minimum	Not fixed	—	Passes by the SC 2.5.8 inline-in-sentence exception

Checked and clean. Zero horizontal overflow at 360, 390, 430 and 820 pixels. Every internal link resolves. Contrast on Paper: Ink 16.88:1, Signal 11.18:1, Caution 4.76:1 — all above the 4.5:1 AA threshold for body text.

What a tablet hid. At 820px the chart text rendered at 7.2px and the nav sat on one line. Neither looked broken. The nav wraps at any width below 540px, so findings 1 and 2 are only visible on a phone.

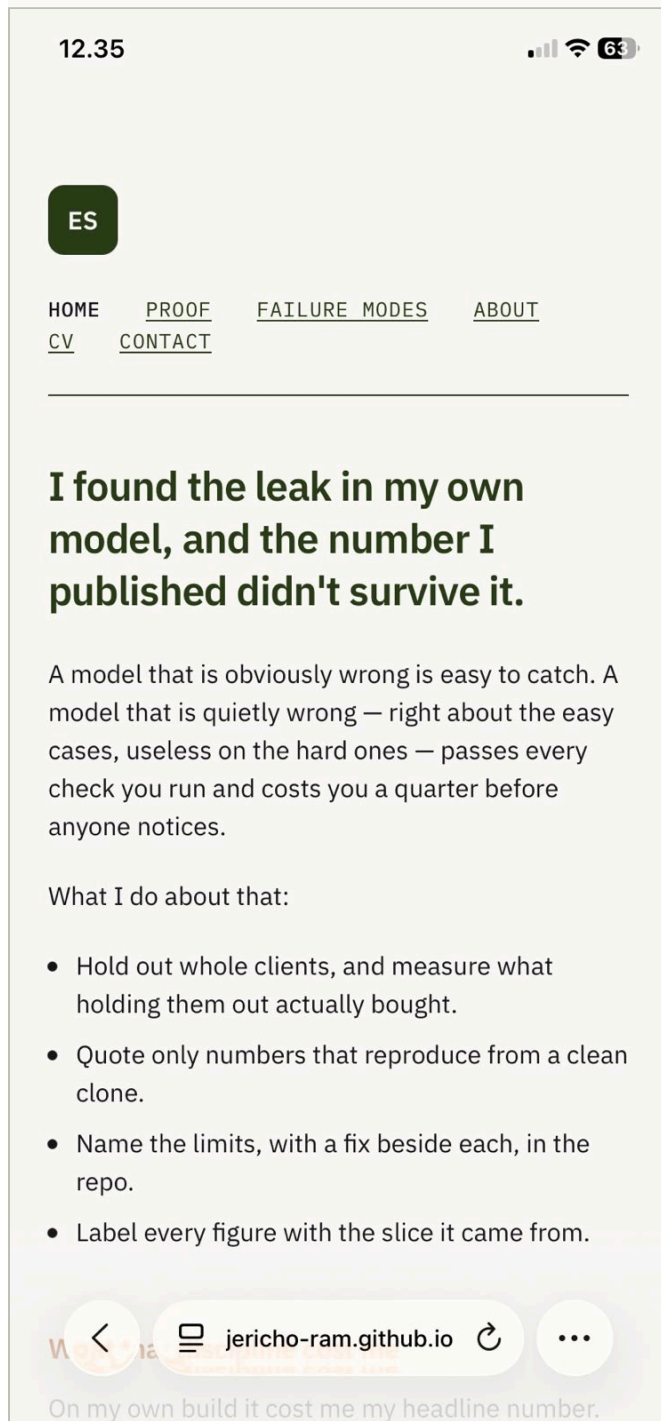
Finding 6 is a repeat. The same shape appeared in the model repo: a retraction notice lived in outputs/model_report.md but not in the script that generates it, so re-running the script would have silently unretracted it. There the fix was to embed the claim in the generator. Here there is no generator — the caption is hand-written — so the binding is a habit rather than a mechanism, which is weaker.

Captured on an Android phone in a private window, before and after the push. Private window matters: a cached stylesheet makes a successful deploy look like a failed one. Finding 6 landed at 13:24, after these were taken, so the corrected captions are not visible here.

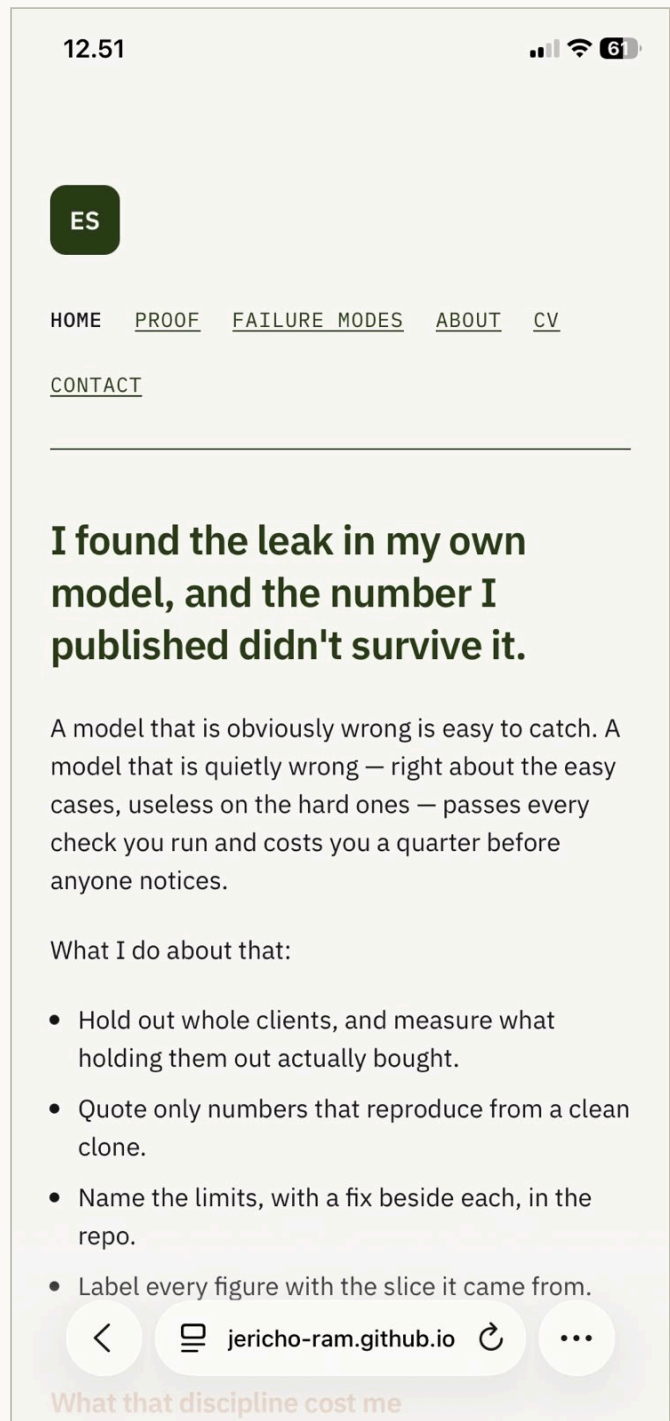
Navigation — finding 2

Before, the two rows sit 20px apart with 20px targets, so FAILURE MODES and CV touch. After, targets are 33px with a 41px row pitch.

BEFORE • 12:35



AFTER • 12:51



Trend distribution chart — findings 1, 3

Before, label text renders at 4.2px and the caption prose is set in mono. After, 12.1px and Plex Sans.

BEFORE · 12:35

12.35

63

01 · The leak that carried the result

Of the 30,000 rows in the bundled CSV, 3,388 (11.3%) had zero impressions in the prior 30 days, and every one of them was labelled not-declining — impressions cannot fall below zero. The model was scoring a population that contained thousands of rows whose answer was fixed by arithmetic before any feature was read.

Scoring only the 26,612 pages where decline was arithmetically possible, the random forest's ROC AUC fell from 0.750 to 0.565 and Precision@50 from 0.740 to 0.600, against a base rate of 0.611. The model was doing a job I could have done with a filter. The hard question — which of the pages that could have gone either way actually declined — is the one it could not answer.

Trend direction distribution



Category	Count
down	1.63e+04
stable	5.96e+03
up	4.39e+03

Label distribution, bundled CSV, the 26,612 pages where decline was arithmetically possible — after the zero-impression filter. Bar labels are rounded to three significant figures by the chart script. Source: outputs/charts/trend_distribution.svg, recoloured, data unchanged.

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AFTER · 12:51

12.51

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The number that didn't reproduce

Feature importance chart — findings 1, 3, 4

The clipping is legible in the before: the left column reads rd_count_tier_1000-2000 and rd_count_tier_2000-3500. Two characters of each name had been cut off the canvas since the chart shipped, on every device.

BEFORE • 12:35

12.35

03 · What I refused to build on

In the March warehouse slice, only 413,966 of 9,841,378 daily rows had GA4 available — 4.2%. The rest were not empty; they were zero-filled, which would have dragged any engagement average toward zero without anything looking wrong on the surface.

So I carry ga4_usable as a 0/1 flag alongside the GA4 features, and the model gets told which is which. At item grain the features are 59% missing, and they are missing because a client had no GA4 that month, not because a page had no engagement. That distinction decides whether a GA4 feature measures behaviour or just identifies a client.

Top model features

log_impressions_90d	0.848
log_clicks_90d	0.588
days_with_impressions	0.43
word_count	0.412
avg_position	0.393
days_since_last_update	0.24
rd_count_tier_1000-2000	0.214
char_count	0.15
days_with_sessions	0.144
rd_count_tier_2000-3500	0.142
age_tier_91-180	0.117
scroll_rate	0.116

Top features by importance. The two largest are both 90d aggregates — the same window flagged in failure mode 3. Source: outputs/charts/top_feature_importance.svg, recoloured, data unchanged.

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AFTER • 12:52

12.52

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